



ANJALAI AMMAL MAHALINGAM ENGINEERING COLLEGE

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING**

EDUCATIONAL ORGANIZATIONAL USING SERVICENOW

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Team Size : 4

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PROBLEM STATEMENT:

Educational institutions often struggle with managing large volumes of student and staff information through manual or disconnected systems. Tasks such as admissions, student record management, and progress tracking become time-consuming and prone to errors. The lack of a unified digital platform leads to inefficiency, communication gaps, and delays in decision-making.

This project addresses these challenges by developing an **Educational Management System** on the **ServiceNow platform** to automate administrative workflows, centralize data, and improve institutional efficiency.

OBJECTIVES:

To design and implement a centralized system for managing student and teacher information using ServiceNow.

1. To automate key processes such as admissions, attendance, and progress monitoring.
2. To enable real-time access and updates of academic and administrative data.
3. To improve coordination among departments and reduce manual workloads.
4. To generate reports and analytics for better academic and administrative decision-making.
5. To enhance transparency and user experience for students, faculty, and administrators.

Skills and Technologies Used:

Platform: ServiceNow

Skills:

- ServiceNow Administration and Configuration
- Form and Table Design
- Workflow and Flow Designer
- Reporting and Dashboard Creation

Task Initiation:

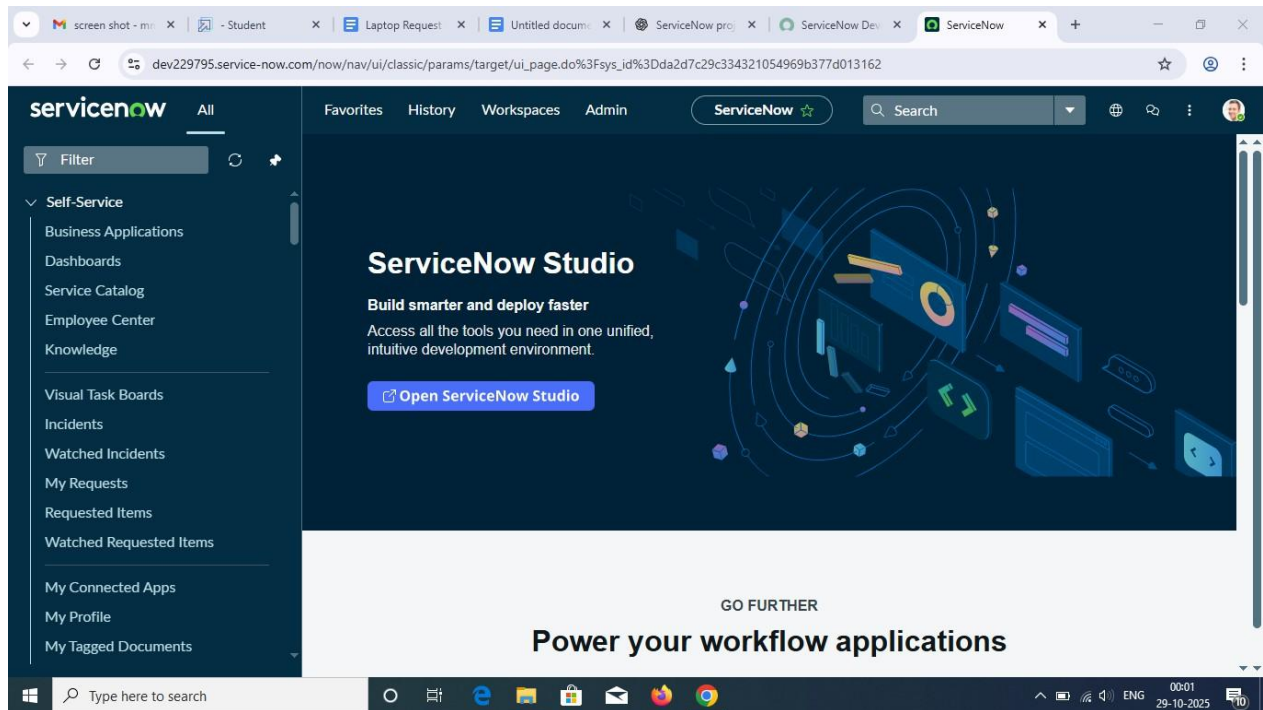
The **Educational Organization Using ServiceNow** project begins with the initiation phase, where the project scope, goals, and resources are clearly defined. The purpose of this phase is to establish a clear understanding of what the project aims to achieve and to prepare the foundation for successful implementation.

Features:

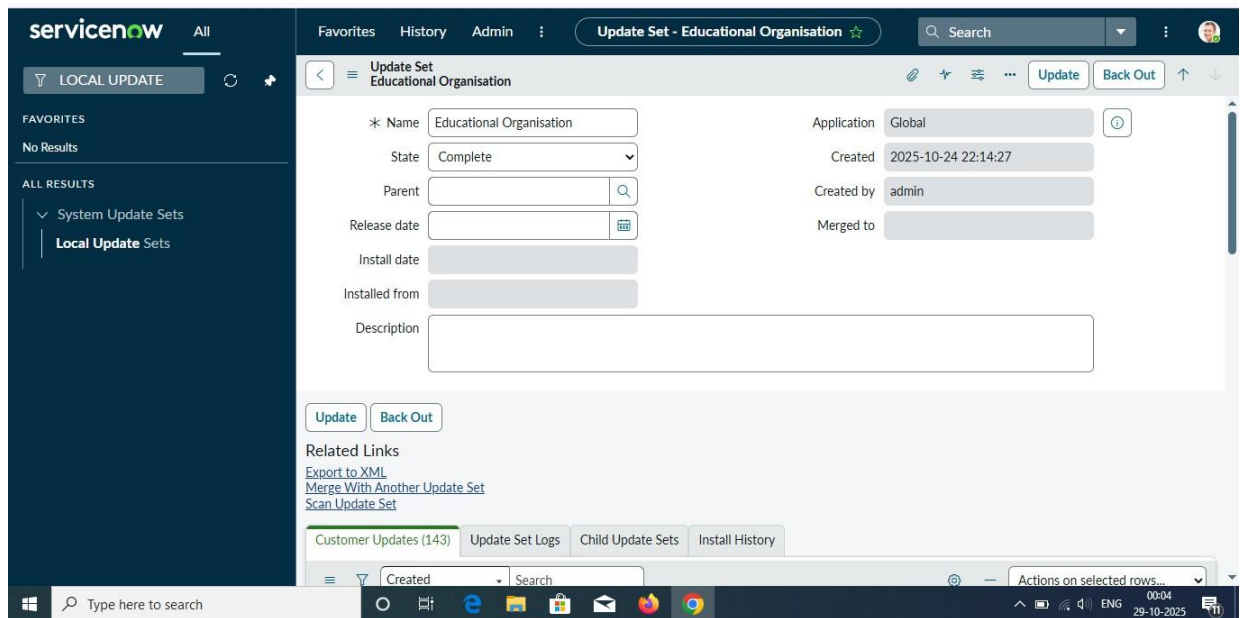
1. Centralized management of student and teacher information on a single ServiceNow platform.
2. Automated workflows for admissions, attendance, and progress tracking.
3. Real-time data access with reporting and dashboard analytics.
4. User-friendly interface for administrators, faculty, and students to improve efficiency.

STEPS:

STEP 1: Setting up ServiceNow Instance



STEP 2: Creating An Update Set



STEP 3: Creating A Table

Salesforce:

The screenshot shows the ServiceNow interface for configuring a table named 'Salesforce'. The left sidebar contains a navigation menu with options like 'Tables', 'Table Rotations', 'Decision Tables', and 'Remote Tables'. The main content area is titled 'Table - Salesforce' and includes a description: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'. Below the description, there are input fields for 'Label' (Salesforce), 'Name' (u_salesforce), 'Application' (Global), and 'Remote Table'. The 'Columns' tab is selected, showing a list of 'Dictionary Entries' with columns: Column label, Type, Reference, Max length, Default value, and Display. The entries include Admin Number, Class, Created, Created by, Grade, Number, Sys ID, Sys ID (GUID), Updated, Updated by, and Updates.

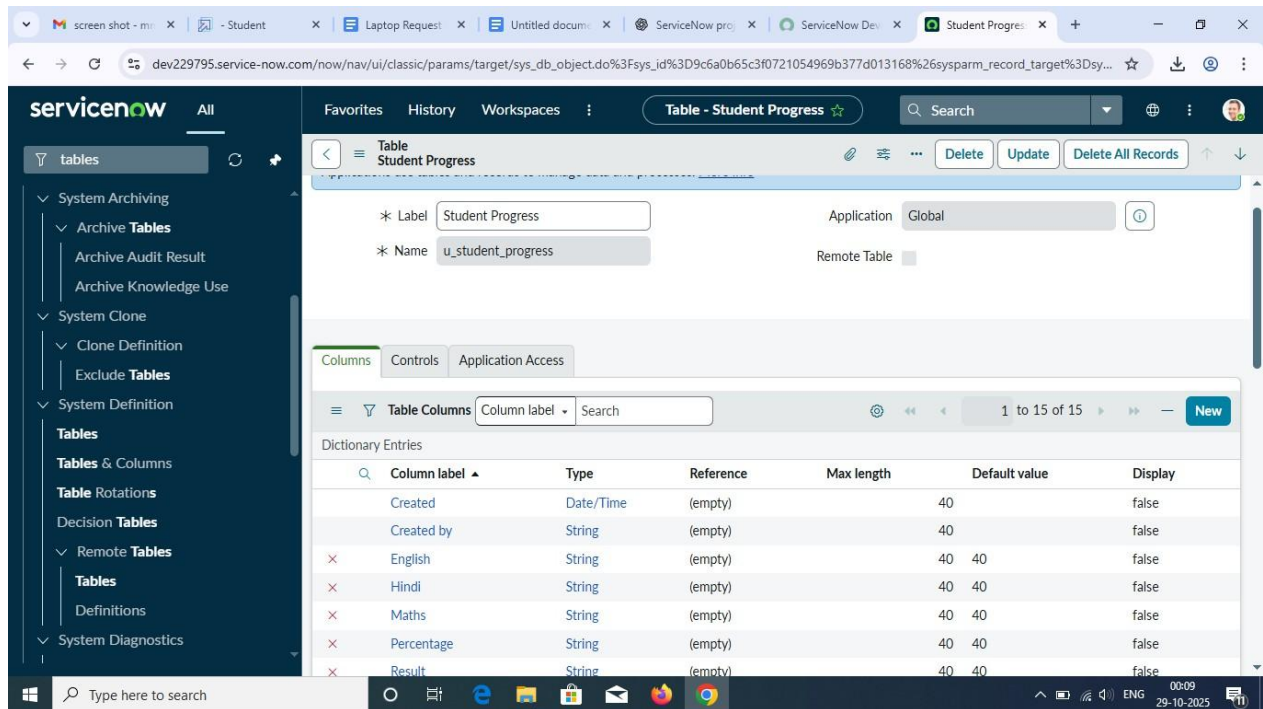
Column label	Type	Reference	Max length	Default value	Display
Admin Number	Integer	(empty)	40		true
Class	System Class Name	(empty)	80	javascript:current.getTableName();	false
Created	Date/Time	(empty)	40		false
Created	Date/Time	(empty)	40	40	false
Created by	String	(empty)	40		false
Created by	String	(empty)	40	40	false
Grade	Choice	(empty)	40		false
Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Sys ID	Integer	(empty)	40	32	false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Updated by	String	(empty)	40	40	false
Updates	Integer	(empty)	40		false
Updates	Integer	(empty)	40	40	false

Admission:

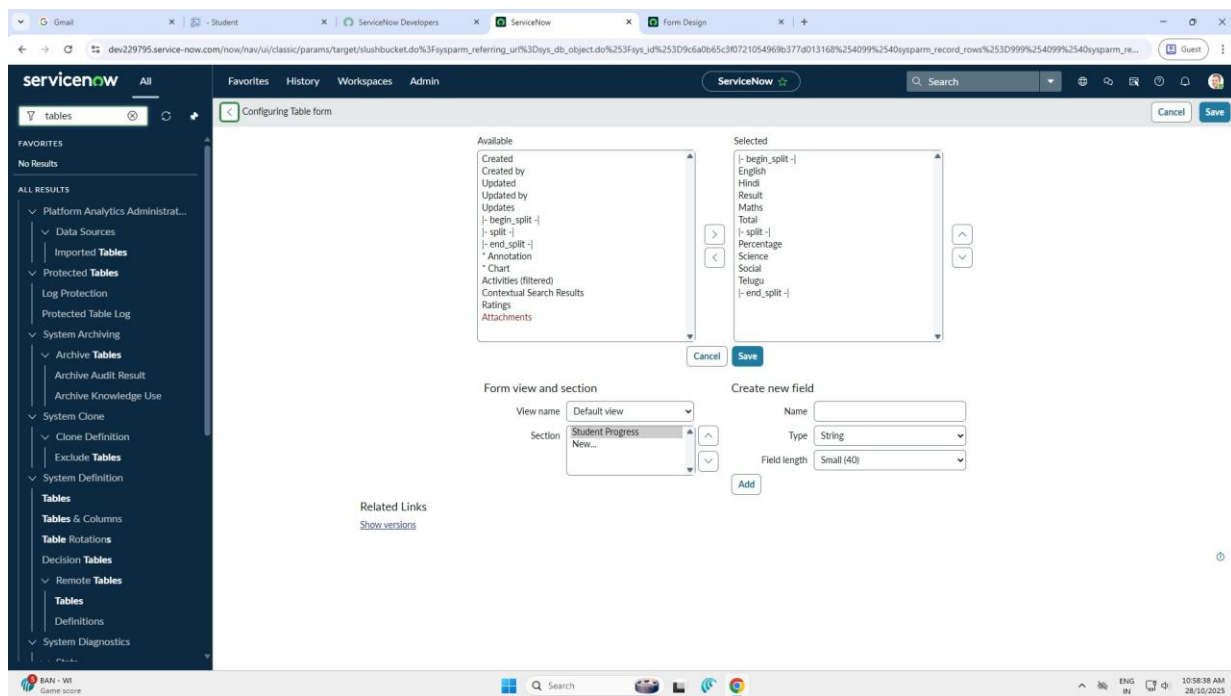
The screenshot shows the ServiceNow interface for configuring a table named 'Admission'. The left sidebar contains a navigation menu with options like 'System Archiving', 'Archive Tables', 'System Clone', 'Clone Definition', 'Exclude Tables', 'System Definition', 'Tables', 'Table Rotations', 'Decision Tables', 'Remote Tables', 'Definitions', and 'System Diagnostics'. The main content area is titled 'Table - Admission' and includes a description: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'. Below the description, there are input fields for 'Label' (Admission), 'Name' (u_admission), 'Application' (Global), 'Remote Table', and 'Extends table' (Salesforce). The 'Columns' tab is selected, showing a list of 'Dictionary Entries' with columns: Column label, Type, Reference, Max length, Default value, and Display. The entries include Admin Number, Admin Status, Area, City, and Class.

Column label	Type	Reference	Max length	Default value	Display
Admin Number	Integer	(empty)	40		true
Admin Status	Choice	(empty)	40	40	false
Area	String	(empty)	40	40	false
City	String	(empty)	40	40	false
Class	System Class Name	(empty)	80	javascript:current.getTableName();	false

Student Progress:



STEP 4: Form Layout



STEP 5: Form Design

The screenshot shows the ServiceNow Form Design interface for the 'Admission' table. The left sidebar contains a 'Fields' list with various attributes like Admin Number, Area, City, Class, Comments, Created, District, House No, Mandal, Updated, and Updates. Below this is a 'Formatters' section with options like Activities (filtered), Contextual Search Results, Process Flow, and Ratings. The main area displays the 'Admission [u_admission]' form layout, which is a vertical stack of fields: Created(u_created), Updated by(u_updated_by), Sys ID, Created by(u_created_by), Updates(u_updates), Admin Status, Purpose of join, Number, School Area, Created by, Grade, Fee, Updated by, School, and Pincode. Each field has a small icon to its right for configuration. The top of the interface shows the 'Form Design' tab and 'Undo'/'Save' buttons. The bottom status bar indicates the user is 'BAN - WI' and the date is '28/10/2025'.

STEP 6: Number Maintenance

Creating Number Maintenance for Admin Number

servicenow

AllFavoritesHistoryWorkspacesAdmin

Table - Salesforce

Search

UpdateDeleteDelete All Records

number

FAVORITESNo Results

ALL RESULTS

System Definition

Number Maintenance

	Class Name	Reference	Max length	Default value	Display
Created by	String	(empty)	80	javascript:current.getTableName();	false
Created	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updates	Integer	(empty)	40		false
Updated by	String	(empty)	40		false

Fill the details >> Submit.

servicenow All Favorites History Workspaces Admin Number - SAL

Number SAL

* Table Salesforce

Prefix SAL

* Number 1,000

Application Global

Number of digits 7

Update Delete

Related Links
[Show Counter](#)

STEP 7: Process Flow

1. Creating Process Flow for Admission Table

- All >> Process Flow>> New.
- Fill the Details as given Below

Flow Formatter New

* Table Admission [u_admission]

* Name New

Application Global

* Label New

Order 1

Active ☒

Condition Add Filter Condition Add "OR" Clause

Admin Status is New

Description

Update Delete

- Right Click on toggle and click on the save .
- Replace the Name and Label as below and click on Insert on stay.

Flow Formatter In progress

* Table Admission [u_admission]

* Name In progress

Application Global

* Label In progress

Order 2

Active ☒

Condition Add Filter Condition Add "OR" Clause

Admin Status is Join in progress

Description

Update Delete

- Replace the Name and Label in order and click on Insert on stay.
Joined >> Rejected >> Rejoined >> Closed >> Cancelled.
- Order should be New >> InProgress >> Joined >> Rejected >> Rejoined >> Closed >> Cancelled.

STEP 8: Client Script

1. Creating “Auto populate” Client Scripts for Admission Table

- All >> Client Scripts >> New.
- Fill the Details as given.

This form has annotations - click ⓘ to toggle them - ([click here](#) to never show this again)

Name: Auto populate

Table: Admission [u_admission]

UI Type: Mobile / Service Portal

Type: onChange

Field name: Admin Number

Application: Global ⓘ

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script

```

1 function onChange(control, oldValue, newValue, isLoading, isTemplate) {
2   if (isLoading || newValue === '') {
3     return;
4   }
5   //Type appropriate comment here, and begin script below
6
7
8 }

```

- Write the Code as below, Enable Isolate script and Save.

```

function onChange(control, oldValue, newValue, isLoading, isTemplate) {
  if (isLoading || newValue === '') {
    return;
  }
}

```

2. Creating “Pincode Update” Client Scripts for Admission Table

- Fill the Details as given.

Client Script
Pincode Update

This form has annotations - click ⓘ to toggle them - (click here to never show this again)

Name: Pincode Update

Table: Admission [u_admission]

UI Type: Desktop

Type: onChange

Field name: Pincode

Application: Global ⓘ

Active: ☒

Inherited: ☐

Global: ☒

Description:

Messages:

Script ⓘ

```

1 function onChange(control, oldValue, newValue, isLoading, isTemplate) {
2   if (isLoading || newValue === '') {
3     return;
4   }
5   var a = g_form.getValue('u_pincode');
6   if(a == '509358')
7   {
8     g_form.setValue('u_mandal', 'kadthal');
9     g_form.setValue('u_city', 'kadthal');

```

- Write the Code as below, Enable Isolate script and Save.

```

function onChange(control, oldValue, newValue, isLoading, isTemplate) {
  if (isLoading || newValue === '') {
    return;
  }
  var a = g_form.getValue('u_pincode');
  if(a == '509358')
  {
    g_form.setValue('u_mandal', 'kadthal');
    g_form.setValue('u_city', 'kadthal');
    g_form.setValue('u_district', 'RangaReddy');

  }
  else if(a == '500081')
  {
    g_form.setValue('u_mandal', 'karmanghat');
    g_form.setValue('u_city', 'karmanghat');
    g_form.setValue('u_district', 'RangaReddy');

  }
  else if(a == '500079')
  {
    g_form.setValue('u_mandal', 'Abids');
    g_form.setValue('u_city', 'AsifNagar');
    g_form.setValue('u_district', 'Hyderabad');

```

```
}
```

```
//Type appropriate comment here, and begin script below
```

```
}
```

3. Creating “Disable Fields” Client Scripts for Student progress Table

- Fill the Details as given.

The screenshot shows the 'Client Script' configuration window for a script named 'Disable Fields'. The window has a title bar with a back arrow, a menu icon, the text 'Client Script Disable Fields', and buttons for 'Update' and 'Delete'. Below the title bar is a blue notification bar that says 'This form has annotations - click ⓘ to toggle them - (click here to never show this again)'. The main form contains several fields: 'Name' (Disable Fields), 'Table' (Student Progress [u_student_progress]), 'UI Type' (All), 'Type' (onLoad), 'Application' (Global), 'Active' (checked), 'Inherited' (unchecked), and 'Global' (checked). There are also empty text areas for 'Description' and 'Messages'. At the bottom is a 'Script' section with a toolbar and a code editor containing the following code:

```
1 function onLoad() {  
2     //Type appropriate comment here, and begin script below  
3     g_form.setDisabled('u_total',true);  
4     g_form.setDisabled('u_percentage',true);  
5     g_form.setDisabled('u_result',true);  
6 }
```

- Write the Code as below, Enable Isolate script and Save.

```
function onLoad() {
```

```
//Type appropriate comment here, and begin script below
```

```
g_form.setDisabled('u_total',true);
```

```
g_form.setDisabled('u_percentage',true);
```

```
g_form.setDisabled('u_result',true);
```

```
}
```

4. Creating “Total Update” Client Scripts for Student progress Table

- Fill the Details as given.

The screenshot shows the ServiceNow interface for configuring a Client Script. The script is named 'Total Update' and is associated with the 'Student Progress [u_student_progress]' table. The UI Type is set to 'All', the Type is 'onChange', and the Field name is 'Social'. The Application is 'Global'. The script is active and inherited. The script code is as follows:

```

function onChange(control, oldValue, newValue, isLoading, isTemplate) {
    if (isLoading || newValue === '') {
        return;
    }
    //Type appropriate comment here, and begin script below
    if (newValue){
        var a = parseInt(g_form.getValue('u_telugu'));
        var b = parseInt(g_form.getValue('u_hindi'));
        var c = parseInt(g_form.getValue('u_english'));
        var d = parseInt(g_form.getValue('u_maths'));
        var e = parseInt(g_form.getValue('u_science'));
        var f = parseInt(g_form.getValue('u_social'));
        var Total = parseInt(a+b+c+d+e+f);
        g_form.setValue('u_total', Total);
    }
}

```

The 'Isolate script' checkbox is checked.

- Write the Code as below, Enable Isolate script and Save.

```

function onChange(control, oldValue, newValue, isLoading, isTemplate) {
    if (isLoading || newValue === '') {
        return;
    }
}

```

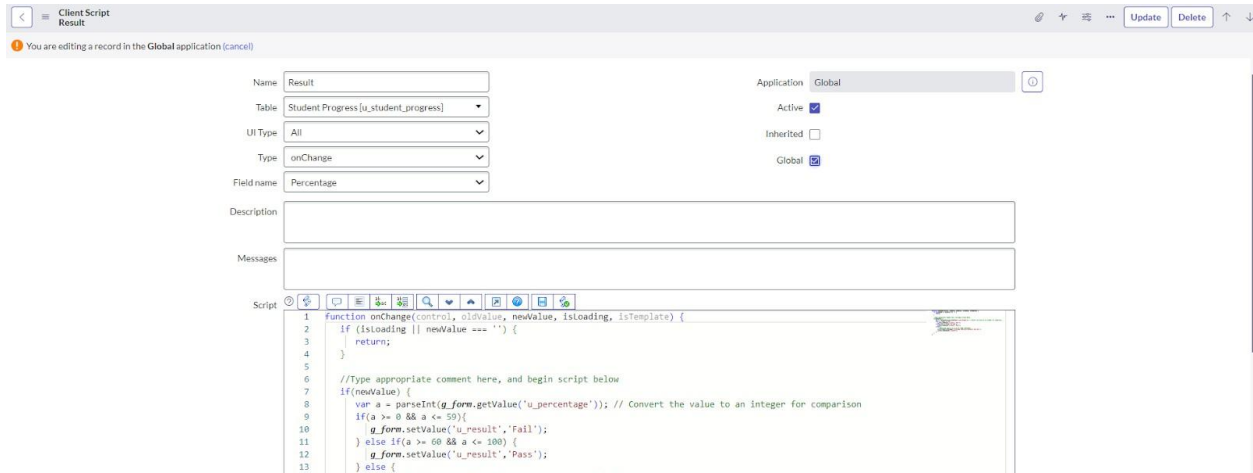
```

//Type appropriate comment here, and begin script below
if (newValue){
    var a = parseInt(g_form.getValue('u_telugu'));
    var b = parseInt(g_form.getValue('u_hindi'));
    var c = parseInt(g_form.getValue('u_english'));
    var d = parseInt(g_form.getValue('u_maths'));
    var e = parseInt(g_form.getValue('u_science'));
    var f = parseInt(g_form.getValue('u_social'));
    var Total = parseInt(a+b+c+d+e+f);
    g_form.setValue('u_total', Total);
}
}

```

Creating “Result” Client Scripts for Student progress Table

- Fill the Details as given.



- Write the Code as below, Enable Isolate script and Save.

```

function onChange(control, oldValue, newValue, isLoading, isTemplate) {
  if (isLoading || newValue === '') {
    return;
  }
  //Type appropriate comment here, and begin script below
  if(newValue) {
    var a = parseInt(g_form.getValue('u_percentage')); // Convert the value to an
integer for comparison
    if(a >= 0 && a <= 59){
      g_form.setValue('u_result','Fail');
    } else if(a >= 60 && a <= 100) {
      g_form.setValue('u_result','Pass');
    } else {
      // Handle the case if a is out of range (optional)
      g_form.addErrorMessage('Percentage should be between 0 and 100.');
```

5. Creating “Percentage” Client Scripts for Student progress Table

- Fill the Details as given.

The screenshot shows the ServiceNow Client Script configuration interface. The 'Name' field is 'Percentage'. The 'Table' is 'Student Progress [u_student_progress]'. The 'UI Type' is 'All'. The 'Type' is 'onChange'. The 'Field name' is 'Total'. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Inherited' checkbox is unchecked. The 'Global' checkbox is checked. The 'Description' field is empty. The 'Messages' field is empty. The 'Script' field contains the following code:

```
1 function onChange(control, oldValue, newValue, isLoading, isTemplate) {
2   if (isLoading || newValue === '') {
3     return;
4   }
5
6   //Type appropriate comment here, and begin script below
7   var Total = g_form.getValue('u_total');
8   var Percentage = (Total/600)*100;
9   g_form.setValue('u_percentage', Percentage+'%');
10 }
```

The 'Isolate script' checkbox is checked. The 'Update' and 'Delete' buttons are visible at the bottom.

- Write the Code as below, Enable Isolate script and Save.

```
function onChange(control, oldValue, newValue, isLoading, isTemplate) {
  if (isLoading || newValue === '') {
    return;
  }
}
```

```
//Type appropriate comment here, and begin script below
var Total = g_form.getValue('u_total');
var Percentage = (Total/600)*100;
g_form.setValue('u_percentage', Percentage+'%');
}
```

Result:

The ServiceNow-based Educational Management System successfully demonstrates the automation of student admission, academic progress, and information management. Data is centralized, workflows are simplified, and administrators can easily generate reports and monitor academic progress.

<
Admission New record

Submit

New	In progress	Joined	Rejected	Rejoined	Closed	Cancelled
-----	-------------	--------	----------	----------	--------	-----------

Admission Number

Purpose of join

Student Name

Father Name

Mother Name

Comments

Admin Date

Grade

Fee \$

Father Cell

Mother Cell

Admin Status

School Details Address

School Area

School

This project stands as a practical example of how modern cloud platforms like **ServiceNow** can revolutionize educational management systems, paving the way for a smarter and more connected academic environment.

